q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Program to calculate total purchase cost including discount and tax

# Input
prices = list(map(float, input("Enter item prices separated by space: ").split()))
quantities = list(map(int, input("Enter item quantities separated by space: ").split()))
discount_rate = float(input("Enter discount rate (%): "))
tax_rate = float(input("Enter tax rate (%): "))

# Processing
subtotal = 0
for i in range(len(prices)):
    subtotal += prices[i] * quantities[i]

discount = subtotal * (discount_rate / 100)
after_discount = subtotal - discount
tax = after_discount * (tax_rate / 100)
total_cost = after_discount + tax

# Output
print("Subtotal:", subtotal)
print("Discount Amount:", discount)
print("Tax Amount:", tax)
print("Total Cost:", total_cost)
```

Ln: 19 Col: 8

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
Enter item prices separated by space: 12 02 42 12
Enter item quantities separated by space: 2 5 4 1
Enter discount rate (%): 10
Enter tax rate (%): 12
Subtotal: 214.0
Discount Amount: 21.400000000000002
Tax Amount: 23.112
Total Cost: 215.712
>>>
```

Ln: 13 Col: 0

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import pandas as pd

order_data = pd.DataFrame({
    'CustomerID': [1, 1, 2, 2, 3],
    'OrderDate': pd.to_datetime(['2024-01-01','2024-01-05','2024-01-03','2024-01-10','2024-01-08']),
    'Product': ['A','B','A','B','A'],
    'Quantity': [2, 3, 1, 5, 4]
})

orders_per_customer = order_data.groupby('CustomerID').size()
avg_quantity_product = order_data.groupby('Product')['Quantity'].mean()
earliest_date = order_data['OrderDate'].min()
latest_date = order_data['OrderDate'].max()

print(orders_per_customer)
print(avg_quantity_product)
print(earliest_date, latest_date)
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
CustomerID
1    2
2    2
3    1
dtype: int64
Product
A    2.333333
B    4.000000
Name: Quantity, dtype: float64
2024-01-01 00:00:00 2024-01-10 00:00:00
>>>
```

Ln: 15 Col: 0

Ln: 1 Col: 19

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import pandas as pd

sales_data = pd.DataFrame({
    'Product': ['Pen', 'Book', 'Bag'],
    'Price': [10, 50, 300],
    'Quantity': [5, 2, 1]
})

sales_data['Total Sales'] = sales_data['Price'] * sales_data['Quantity']
print(sales_data)
```

Ln: 11 Col: 0

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
  Product  Price  Quantity  Total Sales
0     Pen     10         5           50
1    Book     50         2          100
2     Bag    300         1          300
>>>
```

Ln: 9 Col: 0

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import pandas as pd

property_data = pd.DataFrame({
    'PropertyID': [101,102,103],
    'Location': ['City','City','Town'],
    'Bedrooms': [3,5,4],
    'Area': [1200,2000,1500],
    'Price': [500000,800000,600000]
})

avg_price = property_data.groupby('Location')['Price'].mean()
more_than_4 = property_data[property_data['Bedrooms'] > 4]
largest_area = property_data.loc[property_data['Area'].idxmax()]

print(avg_price)
print(len(more_than_4))
print(largest_area)
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
Location
City    650000.0
Town    600000.0
Name: Price, dtype: float64
1
PropertyID       102
Location        City
Bedrooms           5
Area            2000
Price         800000
Name: 1, dtype: object
>>>
```

Ln: 16 Col: 0

Ln: 18 Col: 0

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import numpy as np

student_scores = np.array([
    [80, 70, 75, 85],
    [60, 65, 70, 75],
    [90, 85, 88, 92],
    [70, 75, 78, 80]
])

avg_scores = student_scores.mean(axis=0)
subjects = ['Math','Science','English','History']
highest = subjects[np.argmax(avg_scores)]

print(avg_scores)
print("Highest Average Subject:", highest)
```

Ln: 16 Col: 0

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
[75.   73.75 77.75 83.  ]
Highest Average Subject: History
>>>
```

Ln: 7 Col: 0

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import numpy as np

time = np.array([0, 1, 2, 3, 4])
height = np.array([0, 5, 8, 9, 8])
avg_velocity = (height[-1] - height[0]) / (time[-1] - time[0])
print("Average Velocity:", avg_velocity)
```

Ln: 6 Col: 40

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
Average Velocity: 2.0
>>>
```

Ln: 6 Col: 0

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import numpy as np

house_data = np.array([[3,1200,500000],[5,2000,800000],[4,1500,600000]])
avg_price = house_data[house_data[:,0] > 4, 2].mean()
print("Average Sale Price:", avg_price)
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
Average Sale Price: 800000.0
>>>
```

Ln: 6 Col: 0

Ln: 5 Col: 39

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

```python
import numpy as np

prices = np.array([10, 50, 30, 20, 40])
avg_price = prices.mean()
print("Average Price:", avg_price)
```

Ln: 5 Col: 34

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
Average Price: 30.0
>>>
```

Ln: 6 Col: 0

q1.py - C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py (3.13.5)

File Edit Format Run Options Window Help

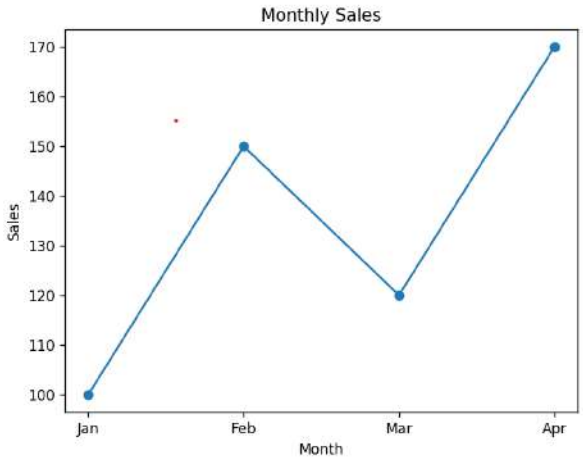
```python
import matplotlib.pyplot as plt

months = ['Jan','Feb','Mar','Apr']
sales = [100,150,120,170]

plt.plot(months, sales, marker='o')
plt.title('Monthly Sales')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.show()
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\bpriy\AppData\Local\Programs\Python\Python313\q1.py ====
```

Figure 1



Ln: 5 Col: 0

Ln: 10 Col: 10